Reference Coordinate Systems (2/3)

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Reference Coordinate Systems

- ECEF Coordinate System
- WGS-84 and PZ-90 Frames
- Geodetic Coordinates
- Geodetic to Cartesian Coordinates
- Cartesian to Geodetic Coordinates
- The Geoid
- Reference Ellipsoids
- Local Geodetic Datums
- Datum Transformations
- Molodensky Transformation
- Rotation Matrices
- ENU Coordinates
- Azimuth & Elevation
- Orthodromes
- Loxodromes

ECEF – Earth-Centered, Earth-Fixed
WGS – World Geodetic System
PZ – Parametry Zemli (Russian)
ENU – East-North-Up

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Reference Ellipsoids

	Reference Ellipsoid Name	Semi-major Axis a (m)	Reciprocal of Flattening $1/f$
	Airy 1830	6377563.396	299.3249646
	Australian National	6378160	298.25
	Bessel 1841 (Ethiopia, Indonesia, Japan and Korea)	6377397.155	299.1528128
	Bessel 1841 (Namibia)	6377483.865	299.1528128
	Clarke 1866	6378206.4	294.9786982
	Clarke 1880	6378249.145	293.465
	Everest (Brunei and E. Malaysia - Sabah and Sarawak)	6377298.556	300.8017
	Everest (India 1830)	6377276.345	300.8017
	Everest (India 1956)	6377301.243	300.8017
	Everest (Pakistan)	6377309.613	300.8017
	Everest (W. Malaysia and Singapore 1948)	6377304.063	300.8017
	Everest (W. Malaysia 1969)	6377295.664	300.8017
GRS-80	Geodetic Reference System 1980	6378137	298.257222101
	Helmert 1906	6378200	298.3
	Hough 1960	6378270	297
	Indonesian 1974	6378160	298.247
Hayford ellipsoid	International 1924	6378388	297
	Krassovsky 1940	6378245	298.3
	Modified Airy	6377340.189	299.3249646
	Modified Fischer 1960	6378155	298.3
	South American 1969	6378160	298.25
PZ-90	Parametry Zemli	6378136	298.257839303
WGS-72	World Geodetic System 1972	6378135	298.26
WGS-84	World Geodetic System 1984	6378137	298.257223563

Sources:
National Imagery and Mapping Agency Technical Report NIMA TR8350.2,
Department of Defense World Geodetic System 1984 – Its Definition and Relationships with Local Geodetic Systems, Third Edition, Amendment 1, 3 January 2000, Appendix A.1
Aeronautical Information Circular of the Russian Federation AIC 04/09, 12.02.2009

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Local Geodetic Datums

Transformation Parameters
Local Geodetic Datums to WGS-84

	Local Geodetic Datum Name	Reference Ellipsoid	Δa (m)	$\Delta f \times 10^4$	No. of Satellite Stations Used	ΔX (m)	ΔY (m)	ΔZ (m)
ED-50	European 1950 (Portugal and Spain)	International 1924	-251	-0.14192702	18	-84 ± 5	-107 ± 6	-120 ± 3
	European 1950 (Mean Solution)	International 1924	-251	-0.14192702	85	-87 ± 3	-98 ± 8	-121 ± 5
NAD-27	North American 1927 (Mean Solution - CONUS)	Clarke 1866	-69.4	-0.37264639	405	-8 ± 5	160 ± 5	176 ± 6
NAD-83	North American 1983 (CONUS)	Geodetic Reference System 1980	0	-0.00000016	216	0 ± 2	0 ± 2	0 ± 2
	Porto Santo 1936 (Porto Santo and Madeira Islands)	International 1924	-251	-0.14192702	2	-499 ± 25	-249 ± 25	314 ± 25
	Sao Braz (Sao Miguel and Santa Maria Islands - Azores)	International 1924	-251	-0.14192702	2	-203 ± 25	141 ± 25	53 ± 25
	Selvagem Grande 1938 (Salvage Islands – Madeira)	International 1924	-251	-0.14192702	1	-289 ± 25	-124 ± 25	60 ± 25

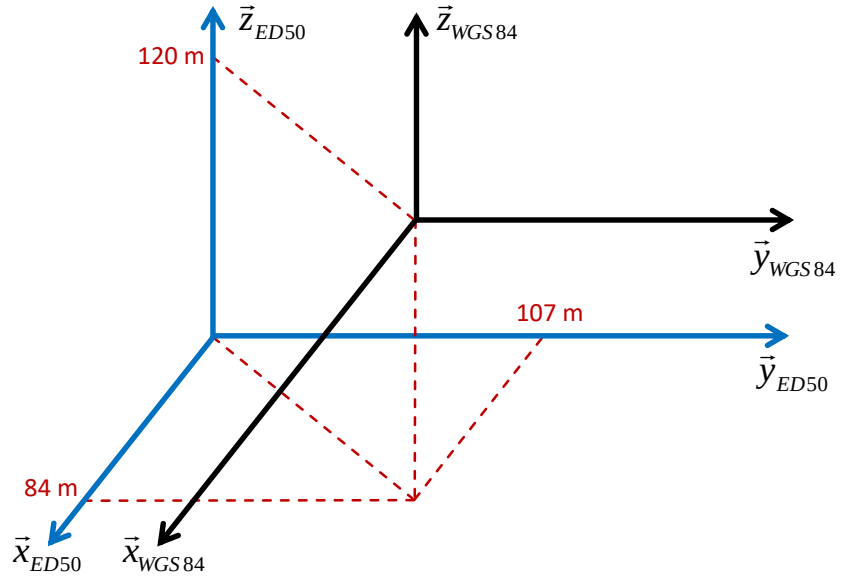
$$P_{\mathrm{WGS84}} = P_{local} + \begin{bmatrix} \Delta X & \Delta Y & \Delta Z \end{bmatrix}^T$$

$$a_{\mathrm{WGS84}} = a_{local} + \Delta a$$

$$f_{\mathrm{WGS84}} = f_{local} + \Delta f$$

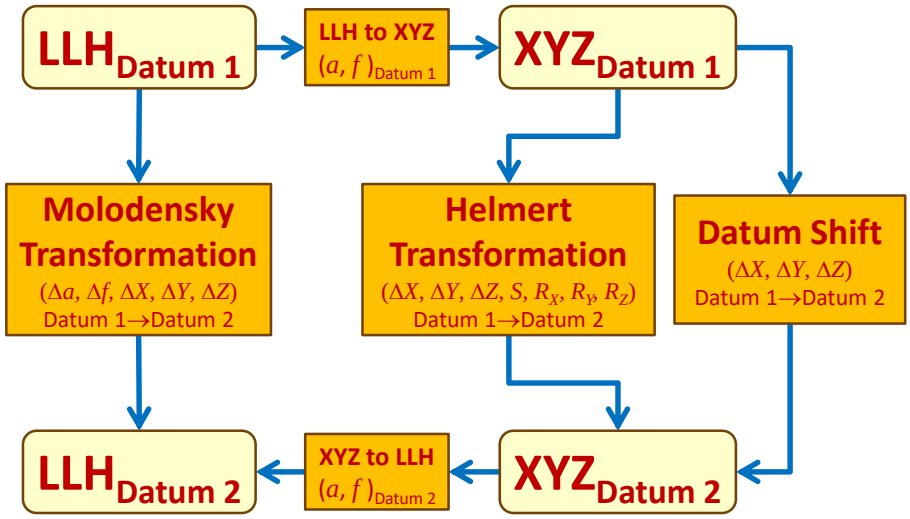
Source:
National Imagery and Mapping Agency Technical Report NIMA TR8350.2,
Department of Defense World Geodetic System 1984 – Its Definition and Relationships with Local Geodetic Systems, Third Edition, Amendment 1, 3 January 2000, Appendix B

ED-50 and WGS-84



European 1950 (Portugal and Spain)

Datum Transformations



low accuracy transformations

Molodensky Transformation

$(\Delta a, \Delta f, \Delta X, \Delta Y, \Delta Z)$
Datum 1→Datum 2

$LLH_{Datum\ 1} \rightarrow LLH_{Datum\ 2}$

$(\phi, \lambda, h, a, b, e, R_N, R_M)_{Datum\ 1}$

$$\Delta\phi_{[radians]} = \frac{-\Delta X \sin\phi\cos\lambda - \Delta Y \sin\phi\sin\lambda + \Delta Z\cos\phi + \left\{\Delta a \frac{R_N e^2}{a} + \Delta f\left[R_M \frac{a}{b} + R_N \frac{b}{a}\right]\right\}\sin\phi\cos\phi}{R_M + h}$$

$$\Delta\lambda_{[radians]} = \frac{-\Delta X \sin\lambda + \Delta Y \cos\lambda}{(R_N + h)\cos\phi}$$

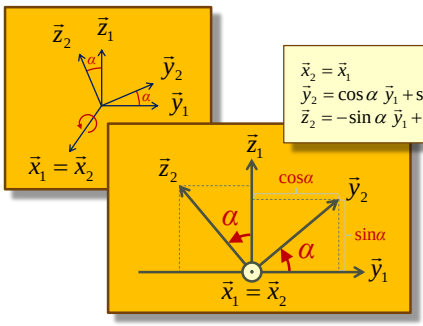
$$\Delta h = \Delta X \cos\phi\cos\lambda + \Delta Y \cos\phi\sin\lambda + \Delta Z \sin\phi - \Delta a \frac{a}{R_N} + \Delta f \frac{b}{a} R_N \sin^2\phi$$

$$\phi_{Datum\ 2} = \phi_{Datum\ 1} + \Delta\phi$$
$$\lambda_{Datum\ 2} = \lambda_{Datum\ 1} + \Delta\lambda$$
$$h_{Datum\ 2} = h_{Datum\ 1} + \Delta h$$

$$R_N = \frac{a}{\sqrt{1 - e^2 \sin^2\phi}}$$

$$R_M = \frac{a(1 - e^2)}{\sqrt{(1 - e^2 \sin^2\phi)^3}}$$

Rotation Matrices



$$\vec{x}_2 = \vec{x}_1$$
$$\vec{y}_2 = \cos\alpha\ \vec{y}_1 + \sin\alpha\ \vec{z}_1$$
$$\vec{z}_2 = -\sin\alpha\ \vec{y}_1 + \cos\alpha\ \vec{z}_1$$

$Rot_x(\alpha)$

$$\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix} = \underbrace{\begin{bmatrix}1 & 0 & 0\\ 0 & \cos\alpha & \sin\alpha\\ 0 & -\sin\alpha & \cos\alpha\end{bmatrix}}\cdot\begin{bmatrix}\vec{x}_1\\ \vec{y}_1\\ \vec{z}_1\end{bmatrix} \Leftrightarrow$$

Rows: coordinates of $\vec{x}_2$, $\vec{y}_2$ and $\vec{z}_2$ in the $(\vec{x}_1,\vec{y}_1,\vec{z}_1)$ ref. system

Columns: coordinates of $\vec{x}_1$, $\vec{y}_1$ and $\vec{z}_1$ in the $(\vec{x}_2,\vec{y}_2,\vec{z}_2)$ ref. system

$$\Leftrightarrow \begin{bmatrix}\vec{x}_1\\ \vec{y}_1\\ \vec{z}_1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\alpha & -\sin\alpha\\ 0 & \sin\alpha & \cos\alpha\end{bmatrix}\cdot\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix} = Rot_x(\alpha)\cdot\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix}$$

$$\Leftrightarrow [x_1\ \ y_1\ \ z_1]\cdot\begin{bmatrix}\vec{x}_1\\ \vec{y}_1\\ \vec{z}_1\end{bmatrix} = \underbrace{[x_1\ \ y_1\ \ z_1]\cdot Rot_x(\alpha)}_{[x_2\ \ y_2\ \ z_2]}\cdot\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix}$$

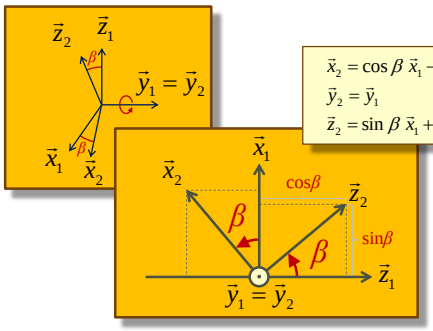
$(\vec{x}_1, \vec{y}_1, \vec{z}_1) \rightarrow Rot_x(\alpha) \rightarrow (\vec{x}_2, \vec{y}_2, \vec{z}_2)$

$$[x_2\ \ y_2\ \ z_2] = [x_1\ \ y_1\ \ z_1]\cdot Rot_x(\alpha)$$

$$Rot_x(\alpha) = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\alpha & -\sin\alpha\\ 0 & \sin\alpha & \cos\alpha\end{bmatrix}$$

Rotation Matrices

$Rot_y(\beta)$



$$\vec{x}_2 = \cos\beta\,\vec{x}_1 - \sin\beta\,\vec{z}_1$$
$$\vec{y}_2 = \vec{y}_1$$
$$\vec{z}_2 = \sin\beta\,\vec{x}_1 + \cos\beta\,\vec{z}_1$$

$$\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix} = \underbrace{\begin{bmatrix}\cos\beta & 0 & -\sin\beta\\ 0 & 1 & 0\\ \sin\beta & 0 & \cos\beta\end{bmatrix}}\cdot\begin{bmatrix}\vec{x}_1\\ \vec{y}_1\\ \vec{z}_1\end{bmatrix} \Leftrightarrow$$

Rows: coordinates of $\vec{x}_2$, $\vec{y}_2$ and $\vec{z}_2$ in the $(\vec{x}_1,\vec{y}_1,\vec{z}_1)$ ref. system

Columns: coordinates of $\vec{x}_1$, $\vec{y}_1$ and $\vec{z}_1$ in the $(\vec{x}_2,\vec{y}_2,\vec{z}_2)$ ref. system

$$\Leftrightarrow \begin{bmatrix}\vec{x}_1\\ \vec{y}_1\\ \vec{z}_1\end{bmatrix} = \begin{bmatrix}\cos\beta & 0 & \sin\beta\\ 0 & 1 & 0\\ -\sin\beta & 0 & \cos\beta\end{bmatrix}\cdot\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix} = Rot_y(\beta)\cdot\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix}x_1 & y_1 & z_1\end{bmatrix}\cdot\begin{bmatrix}\vec{x}_1\\ \vec{y}_1\\ \vec{z}_1\end{bmatrix} = \underbrace{\begin{bmatrix}x_1 & y_1 & z_1\end{bmatrix}\cdot Rot_y(\beta)}_{\begin{bmatrix}x_2 & y_2 & z_2\end{bmatrix}}\cdot\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix}$$

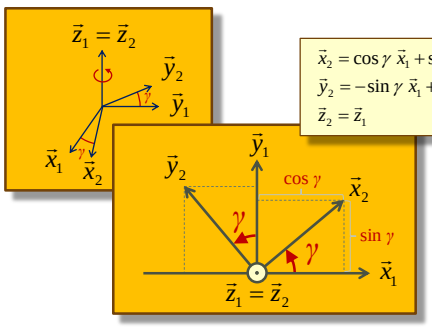
$(\vec{x}_1,\vec{y}_1,\vec{z}_1) \rightarrow Rot_y(\beta) \rightarrow (\vec{x}_2,\vec{y}_2,\vec{z}_2)$

$$\begin{bmatrix}x_2 & y_2 & z_2\end{bmatrix} = \begin{bmatrix}x_1 & y_1 & z_1\end{bmatrix}\cdot Rot_y(\beta)$$

$$Rot_y(\beta) = \begin{bmatrix}\cos\beta & 0 & \sin\beta\\ 0 & 1 & 0\\ -\sin\beta & 0 & \cos\beta\end{bmatrix}$$

Rotation Matrices

$Rot_z(\gamma)$



$$\vec{x}_2 = \cos\gamma\,\vec{x}_1 + \sin\gamma\,\vec{y}_1$$
$$\vec{y}_2 = -\sin\gamma\,\vec{x}_1 + \cos\gamma\,\vec{y}_1$$
$$\vec{z}_2 = \vec{z}_1$$

$$\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix} = \underbrace{\begin{bmatrix}\cos\gamma & \sin\gamma & 0\\ -\sin\gamma & \cos\gamma & 0\\ 0 & 0 & 1\end{bmatrix}}\cdot\begin{bmatrix}\vec{x}_1\\ \vec{y}_1\\ \vec{z}_1\end{bmatrix} \Leftrightarrow$$

Rows: coordinates of $\vec{x}_2$, $\vec{y}_2$ and $\vec{z}_2$ in the $(\vec{x}_1,\vec{y}_1,\vec{z}_1)$ ref. system

Columns: coordinates of $\vec{x}_1$, $\vec{y}_1$ and $\vec{z}_1$ in the $(\vec{x}_2,\vec{y}_2,\vec{z}_2)$ ref. system

$$\Leftrightarrow \begin{bmatrix}\vec{x}_1\\ \vec{y}_1\\ \vec{z}_1\end{bmatrix} = \begin{bmatrix}\cos\gamma & -\sin\gamma & 0\\ \sin\gamma & \cos\gamma & 0\\ 0 & 0 & 1\end{bmatrix}\cdot\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix} = Rot_z(\gamma)\cdot\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix}x_1 & y_1 & z_1\end{bmatrix}\cdot\begin{bmatrix}\vec{x}_1\\ \vec{y}_1\\ \vec{z}_1\end{bmatrix} = \underbrace{\begin{bmatrix}x_1 & y_1 & z_1\end{bmatrix}\cdot Rot_z(\gamma)}_{\begin{bmatrix}x_2 & y_2 & z_2\end{bmatrix}}\cdot\begin{bmatrix}\vec{x}_2\\ \vec{y}_2\\ \vec{z}_2\end{bmatrix}$$

$(\vec{x}_1,\vec{y}_1,\vec{z}_1) \rightarrow Rot_z(\gamma) \rightarrow (\vec{x}_2,\vec{y}_2,\vec{z}_2)$

$$\begin{bmatrix}x_2 & y_2 & z_2\end{bmatrix} = \begin{bmatrix}x_1 & y_1 & z_1\end{bmatrix}\cdot Rot_z(\gamma)$$

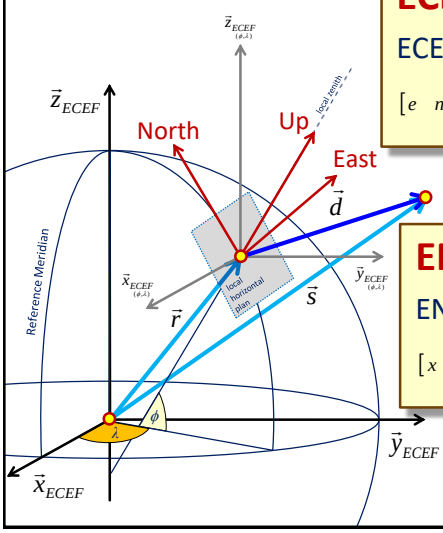
$$Rot_z(\gamma) = \begin{bmatrix}\cos\gamma & -\sin\gamma & 0\\ \sin\gamma & \cos\gamma & 0\\ 0 & 0 & 1\end{bmatrix}$$

Rotation Matrices

Some Properties

- A chain of rotations is still a rotation
- A rotation does not change the vector's size
- $Rot^{-1} = Rot^{T}$

ENU Coordinates



ECEF(ϕ,λ) to ENU

ECEF$_{(\phi,\lambda)}$→Rot$_z$($\lambda+\pi/2$)→Rot$_x$($\pi/2-\phi$)→ENU

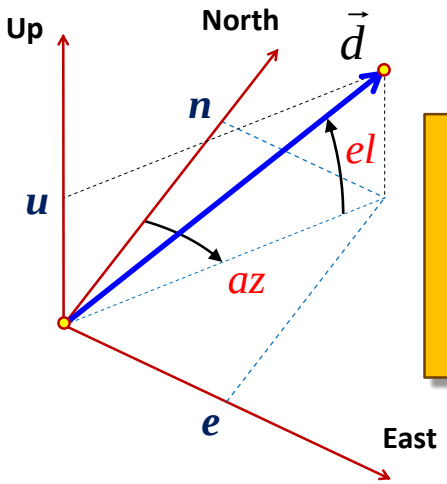
$$[e \quad n \quad u]_{ENU} = [x \quad y \quad z]_{\substack{ECEF \\ (\phi,\lambda)}} \cdot Rot_z\left(\lambda + \frac{\pi}{2}\right) \cdot Rot_x\left(\frac{\pi}{2} - \phi\right)$$

ENU to ECEF(ϕ,λ)

ENU→Rot$_x$($\phi-\pi/2$)→Rot$_z$($-\lambda-\pi/2$)→ECEF$_{(\phi,\lambda)}$

$$[x \quad y \quad z]_{\substack{ECEF \\ (\phi,\lambda)}} = [e \quad n \quad u]_{ENU} \cdot Rot_x\left(\phi - \frac{\pi}{2}\right) \cdot Rot_z\left(-\lambda - \frac{\pi}{2}\right)$$

Azimuth & Elevation



$$az = \arctan\left(\frac{e}{n}\right)$$

$$el = \arctan\left(\frac{u}{\sqrt{n^2 + e^2}}\right)$$

ENU Coordinates

MATLAB hint: use `atan2` to compute *az*